Fragmentation

step 01 : calculate IP payload

4500 − 30 = 4470 B

step 02 : 1500 − 30 = 1470 B

MTU payload $= \left\lfloor \frac{1470}{8} \right\rfloor * 8$

$= 1464$ B

step 03: Calculate DLL payload

step 04 : Number of fragments $\left\lceil \frac{4470}{1464} \right\rceil = 4$

step 05 : Fragment info

IP length = 4500 B

IP header = 30 B

Header	Data

MTU = 1500

	Fragsize	ID	flag ~~length~~	offset
①	1464	x	1	0 → আগের কোন আকার নি
②	1464	x	1	1464/8 = 183
③	1464	x	1	2928/8 = 366
④	78	x	0	4392/8 = 549

4470 − (1464 + 1464 + 1464)

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2.c) MTU = 1500 bytes

Data = 4500 bytes

Actual Data

~~IP payload~~ = 4500 - 30

= 4470

IP payload = 1500 - 30

= 1470

MTU payload = $\left[\frac{1470}{8}\right] * 8$

= ~~1470~~ 1464

No of fragments = $\left\lceil \frac{4470}{1470} \right\rceil$

= 4

	Fragmentsize	flag	ID	offset
①	~~1470~~ 1464	1	x	0/8 = 0
②	1470	1	x	1470/8 =
③	1470	1	x	2940/8 =
④	60	0	x	4410/8 =

Actual Data = 5500 - 20

= 5480

IP Payload = 1516 - 20

= 1496

$\therefore$ MTU = $\left\lfloor \frac{1496}{8} \right\rfloor \times 8$

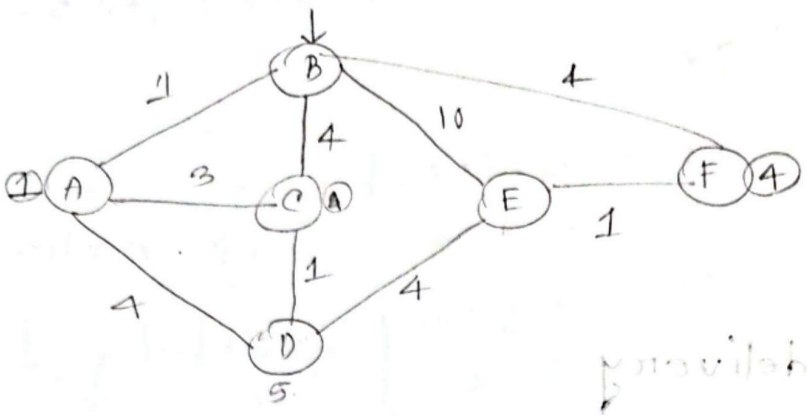
= 1496

$\therefore$ No of segments = $\left\lceil \frac{5480}{1496} \right\rceil$

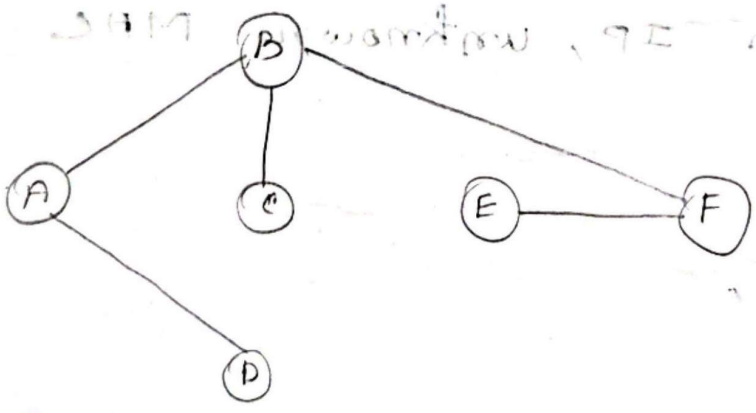
= 4

		ID	flag	offset
①	1496	400	1	0/8 = 0
②	1496	400	1	1496/8 = 187
③	1496	400	1	(1496+1496)/8 = 374
④	992	400	0	1496×3/8 = 561

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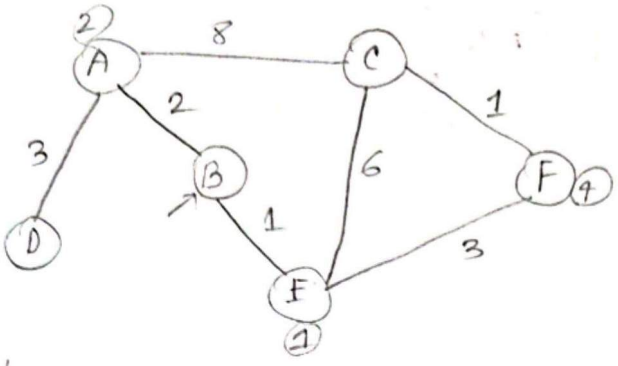
	A	C	D	E	F
init	∞	∞	∞	∞	∞
0 →	1, B	4, B	∞	10, B	4, B
1 →		4, B	5, A	10, B	4, B
2 →			5, A	10, B	4, B
3 →			5, A	5, F	
4 →				5, F	



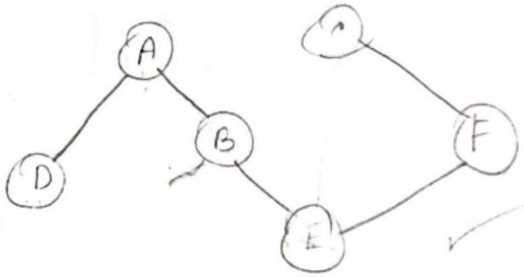
forwarding Table for B :

node	initial	Next link
A	(B, A)	
C	(B, C)	
D	(B, A)	(B, F)
E	(B, F)	

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Iteration	A	C	D	E	F
init	∞	∞	∞	∞	∞
0	2, B	∞	∞	[1, B]	∞
1	[2, B]	7, E	∞		4, E
2		7, E	5, A		[4, E]
3		[5, E]	5, A		
4			[5, A]		



node	Immediate next ~~node~~ link
A	(B, A)
C	(B, E)
D	(B, A)
E	(B, E)
F	(B, E)

ARP steps:

1. check ARP cache Table →

IP	MAC
1.12	BB-BB-

2. ARP Query/Request:

Broad cast

SIP: 192.168.1.10,
DIP: 192.168.1.12
SMAC: AA-AA-AA-AA-AA-AA
DMAC: FF-FF-FF-FF-FF-FF → Broadcast MA

3. ARP Reply/Response:

Unicast

SIP: 1.12
DIP: 1.10
SMAC: BB-BB-BB-;.....
DMAC: AA-AA-AA-....

4. update the ARP table

Router:

1.

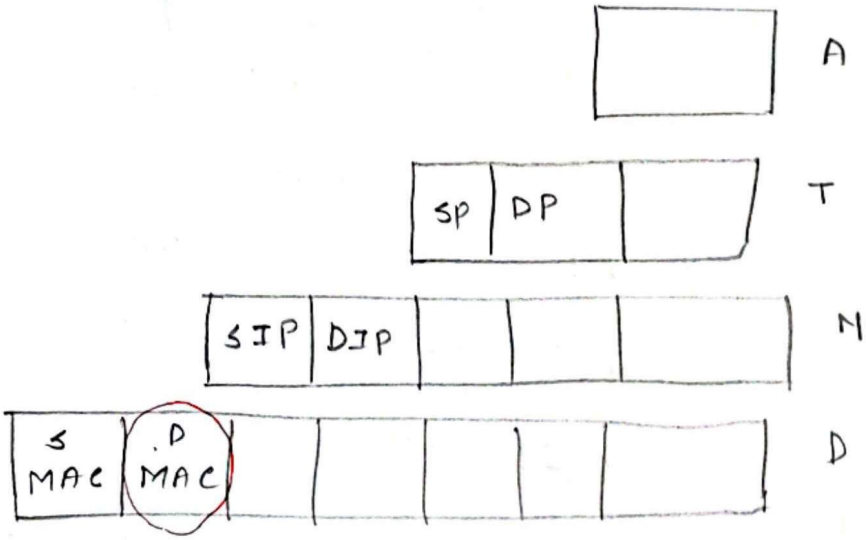
2. SIP: 1.10
DIP: 1.1
SMAC: AA-AA...
DMAC: FF-FF...

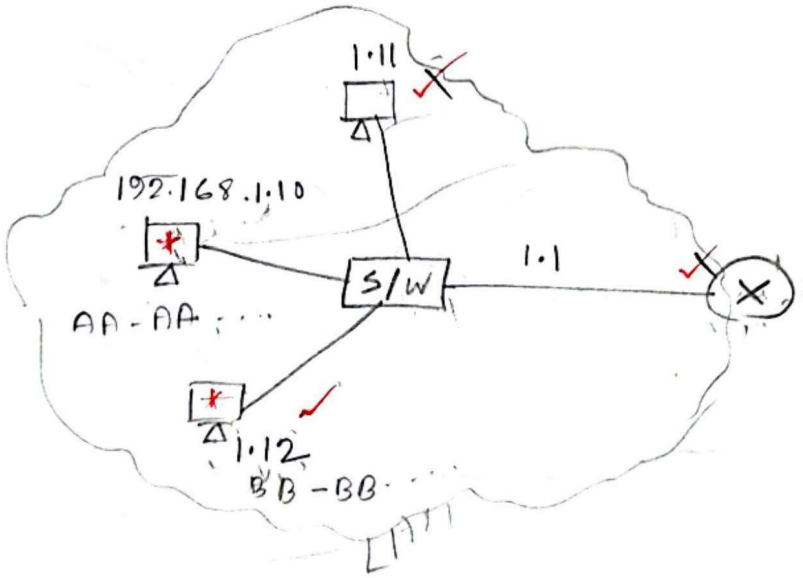
3. SIP: 1.1
DIP: 1.10
SMAC: cc-cc...
DMAC: AA-AA-AA....

Computer Network

Address Resolution Protocol (ARP):

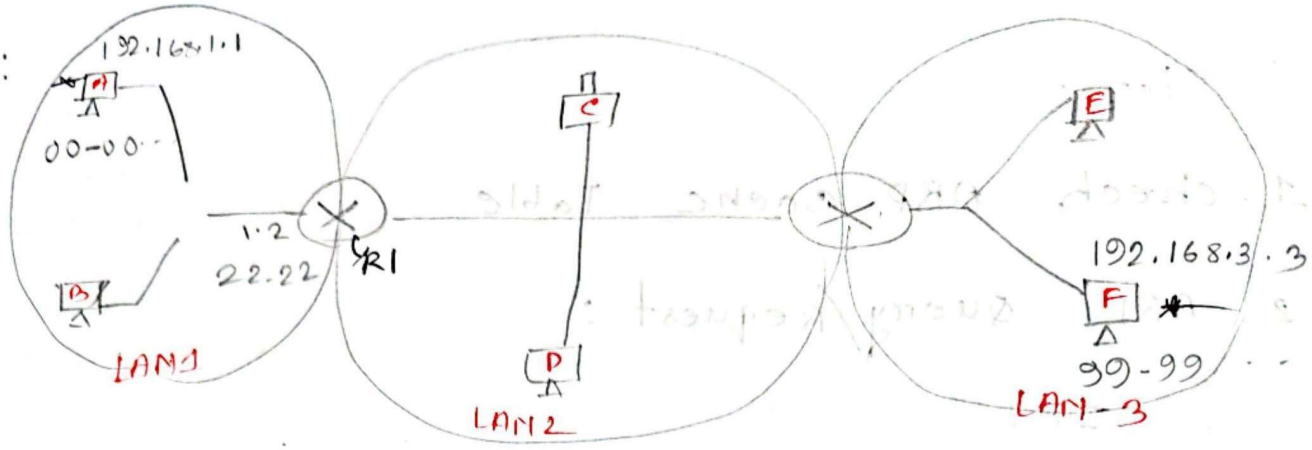
- IP Known
- MAC unknown

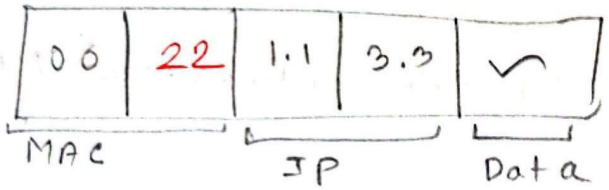




Network এর বাইরে
যাওয়ার জন্য router
এর কাছে যেতে হবে

25):





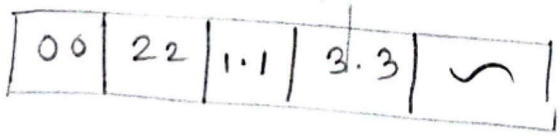
ARP Table

IP	MAC
1.2	22-22-....

(i) FF-FF-FF-FF-FF-FF

(ii) B, Router 1

(iii) Router 1 → 192.168.1.2

(iv) 

MUX: Multiplexing occurs at the senders side The transport Layer collects data from different application processes adds transport layer headers, creates segments and passes them to the network Layer.

DeMux: Demultiplexing occurs at the receiver side. The transport layer examines the head information of incoming segments and delive the data to the appropriate application socket.

Syllabus CN

(1) Transport layer কী

(2) Multiplex — Demux

(3) UDP কীভাবে কাজ করে / UDP vs TCP

(4) Checksum

(5) Reliable Data Transfer
- stop and wait
- Go back N (cumulative)
- selective Repeat (individual)

(6) Tcp seq, ack, retransmission

(7) 3-duplicate

(8) 3 way handshake, 4way handshake
- 3 way handshake → connection stabilish হয় এর জন্য
- 4way handshake → connection terminate করার জন্য

(9) Flow control, congestion control

(10) Congestion control methods.

15 marks

Network Layer:

1. what is ↑?
2. DHCP
3. IP Fragmentation
4. NAT
5. Routing Protocol
   - link state vs distance vector
6. link state simulation
   - ↳ Dijkstra.

—x—

Data link Layer

1. what is ↑?
2. ARP
3. Access control
   - ↳ shared medium
   - → MAC protocol
     - (i) channel partitioning
     - (ii) Random Access
     - (iii) Taking turns.

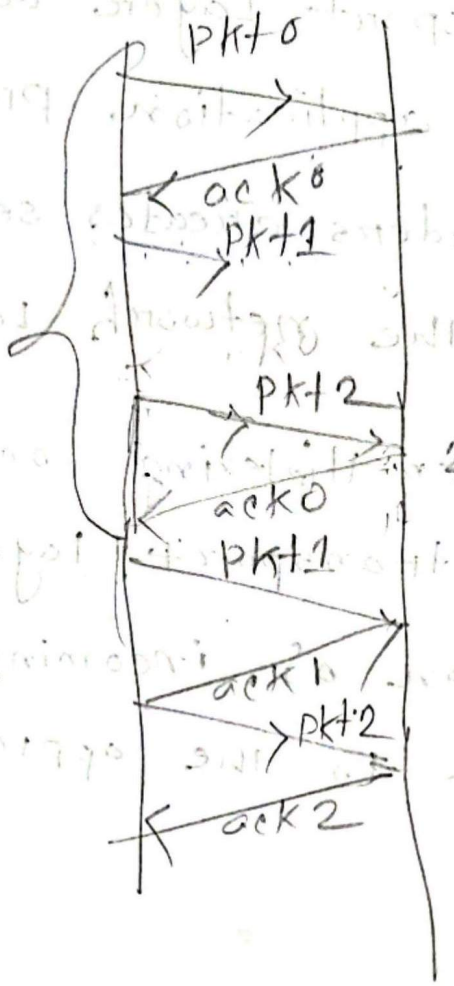
} chapter-6 slide 10, 13, 15

stop and wait



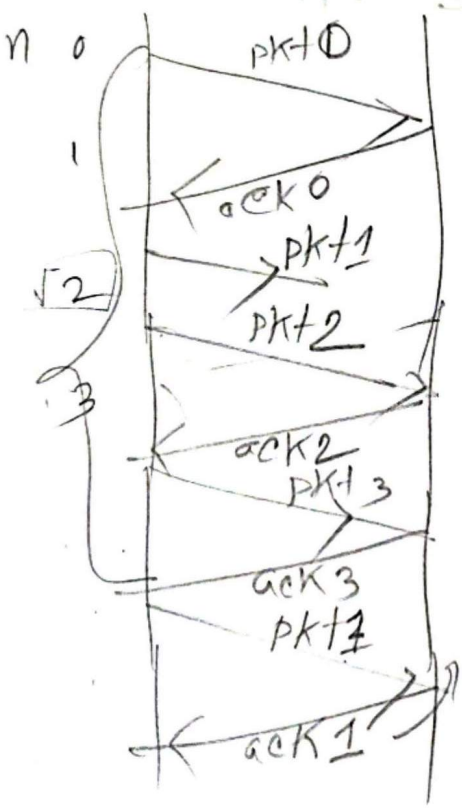
Go-back-N

pkt2 → 0, 1, 2



I love u

Selective (2).



Buffer pkt 2

Buffer. pkt 3

rcv 1
deliver pkt 2, pkt 3.